Research for GenAI and HCI Project

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1

Week 1: Planning and Scoping

- Select the topic and title
- The theme of this project is “Social Equity”
- Using Generative Artificial Intelligence and/or HCI
- However, if students have another topic they find more interesting, you are free to choose it even if it does not directly align with the given theme.

2

Week 2: Literature Review

- Literature review: write draft paper for literature review (English, if possible), Turnitin => เลือกหัวข้อด้านไหน ก็หาเปเปอร์เคสในอดีตแนวนั้น
  - literature review articles = (at least) 6*(the number of members in each group)
  - literature review: Title, Introduction/Background, Related Works, and References
  - e.g., IEEE, Springer, Scopus, Scimago, ScienceDirect, ACM
  - One-third of references should be within 2 years

3

Week 2: Literature Review (cont.)

- In the final paragraph of the Related Works section, there should be a concluding summary that reflects on the papers reviewed.
- Specifically, it should highlight whether the existing methods are sufficient to address the research problem, which aspects are already effective and can be adopted, and which limitations or gaps still require further research or experimentation.
- Additionally, it should clarify how our work differs from the reviewed studies.
- To strengthen this point, a comparative summary table of previous works should be included to emphasize their weaknesses or gaps that our study aims to build upon.

4

Week 3: Research Design

- Select participants & recruitment criteria, if any
- Design methodology
  - Computation, if any
  - Data preparation (check data availability)
  - Training
  - Evaluation -> Novelty
- Design the first experiment/initial prototypes
- Submit a draft paper for literature review to chutisant.ker@nida.ac.th
  - Paper format from https://uist.acm.org/2023/assets/files/word-one-column-submission-sample.docx

5

Week 4: Experiment Setup / Implementation

- Start coding and implementation / Data preparation (clean data, preprocess)
- Build prototypes, if any (Create an **experimental interface** with GenAI integration, e.g., AI chatbot, AI content generator)

6

Week 5: Implementation Progress

- Refine based on initial results/pilot testing
- Collect quantitative/qualitative feedback from users
  - Quantitative: run statistical tests (ANOVA, regression, t-test)
  - Qualitative: thematic coding of interviews / open comments (interviews, focus groups)

7

Week 6: Comparing Results

- Comparing results with state-of-the-art works
- Visualizing results (e.g., table, graph, pie chart)
- Discussion and limitation

8

Week 7 and Week 8

- Week 7: present draft of full paper, Turnitin
  - Full paper (pages) = (at least) 6*(the number of members in each group)
  - Paper format from https://uist.acm.org/2023/assets/files/word-one-column-submission-sample.docx

- Week 8: submit a full paper and final presentation
  - chutisant.ker@nida.ac.th

9

Sample Tools for GenAI Artist

- Midjourney
- https://www.midjourney.com/home/

- DeepDream
- https://deepdreamgenerator.com/

- OpenAI
- https://openai.com/dall-e-2/

- Pinegraph (Still free now)
- https://pinegraph.com/

10

Sample Tools

- Prototyping: Figma, Adobe XD, Sketch → Rapid UI/UX prototyping with mockups
- GenAI models: Hugging Face, OpenAI APIs, Stable Diffusion, Flux
- Questionnaires: SUS, QUIS, TAM, or custom Likert scales
- Think-Aloud Protocol Tools (e.g., Morae, Lookback.io, Validately) → Record user testing sessions
- Agentic AI: n8n (pronounced “n-eight-n”, short for “nodemation”), an open-source workflow automation tool, https://n8n.io/

11

Preparing the Manuscript

- Declaration statement
- Contribution form
  - The role of each author
  - The order in which authors appear
- Two emails of each author

12

The role of each author

- Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing

- Select 2-3 roles and describe linked to the content of the paper

13